

semlex — Manual

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1 Purpose

This document explains how to use the bug report classification tool. The tool evaluates several text classification models for binary issue classification over five datasets: `caffe`, `incubator-mxnet`, `keras`, `pytorch`, and `tensorflow`. The main reported model is a feature-level hybrid that combines TF-IDF features with sentence embeddings.

2 Getting Started

Before using the tool, ensure your environment meets the system and software requirements detailed in `requirements.pdf`. This includes installing the `uv` package manager and ensuring a compatible `LATEX` toolchain is available if you intend to rebuild the report.

From the repository root, initialise the virtual environment with:

```
uv sync
```

Refer to `requirements.pdf` for platform-specific installation instructions and a list of all Python dependencies.

3 Main Commands

3.1 Run the Full Default Experiment

To run the main experiment pipeline across all datasets and regenerate the main report plots:

```
uv run python -m src.run_experiments --with-plots
```

This runs the default preprocessing mode, the main model comparison, the embedding-dimension ablation, the plotting step, and all statistical tests (Wilcoxon and Friedman) automatically.

3.2 Run Every Preprocessing Ablation

To run all preprocessing variants as well as the default pipeline:

```
uv run python -m src.run_experiments --all-preprocessing --with-plots
```

This repeats the experiment workflow for:

- `none`
- `stopwords_all`
- `stopwords_keep_negation`
- `lemmatize`
- `stopwords_keep_negation+lemmatize`

3.3 Run One Dataset Only

For a quicker or targeted run on a single dataset:

```
uv run python -m src.run_experiments --dataset caffe
```

3.4 Run the Baseline Only

To execute only the Naïve Bayes + TF-IDF baseline:

```
uv run python -m src.run_baseline --dataset caffe
```

3.5 Regenerate Plots

If the result CSV files already exist and only the figures need to be rebuilt:

```
uv run python -m src.plot_results
```

3.6 Rebuild the Report

To regenerate the $\text{\LaTeX}$ table fragments and rebuild the main report PDF:

```
uv run python -m src.tools.build_docs --report-only
```

This command regenerates the $\text{\LaTeX}$ table fragments from the current `results/` CSV files and recompiles `docs/report/report.tex`.

4 Outputs

The main outputs are written to `results/`. Important files include:

- per-dataset run outputs such as `results/caffe_all_models_768.csv`
- top-level summary files such as `results/summary_768.csv`
- statistical summary files such as `results/wilcoxon_summary_768.csv`
- ablation summaries such as `results/embedding_ablation_summary_512_256_128_64.csv`
- report figures in `results/figures/`
- generated report notes in `results/report_notes.txt`

Cached embeddings are written to `results/embeddings/`. These caches are separated by dataset, model, dimension, and preprocessing mode, so routine reruns should not require manual cache deletion.

5 Typical Usage

For normal use, the simplest sequence is:

1. ensure dependencies are installed via `uv sync`
2. run the default experiment with `uv run python -m src.run_experiments --with-plots`
3. if needed, run the full preprocessing ablation with `uv run python -m src.run_experiments -all-preprocessing --with-plots`
4. rebuild the report with `uv run python -m src.tools.build_docs --report-only`

6 Troubleshooting

- If the first embedding run is slow, this is expected. The model is being downloaded and cached.
- If `--help` feels slow, use the updated runner scripts. Their heavier imports are deferred until after argument parsing.
- If environment setup or report building fails, refer to `requirements.pdf` to ensure all dependencies and system tools are correctly installed.

7 Artefact Location

`manual.pdf` is placed in the repository root. The source for this PDF is stored in `docs/manual/`.